



Evaluating trends in hospital delirium epidemiology in England between 1998 to 2025

Matthew Luney, MPhil FRCA^{1,2}; Kyle Pattinson, DPhil FRCA¹, Matthew Costa PhD FRCS²
1. Nuffield Dept of Clinical Neurosciences, University of Oxford, UK; 2. Nuffield Dept of Orthopaedics, Rheumatology and Musculoskeletal Sciences, University of Oxford, UK



INTRODUCTION

Acute inpatient care complicated by delirium results in significantly increased morbidity and mortality for patients as well as substantial costs for the healthcare system. There is increasing evidence that a single episode of delirium independently increases a person’s lifetime risk of dementia separate to underlying medical conditions (Gordon 2024).

The organisation of the British National Health Service (NHS) data management and registry reporting of admission data facilitates population level analysis of changes in delirium burden. In order to gain a better understanding of the future potential resource requirements in our healthcare system we describe the trends in the frequency of hospital admissions complicated by delirium in the last 27 years.

METHODOLOGY

The NHS Hospital Episode Statistics data is a registry of all publicly funded in care in NHS and private hospitals in England (Herbert 2017). Admissions of adult inpatients were analysed from registry inception in April 1998 to April 2025. Patients with an episode of delirium were identified from ICD-10 diagnostic codes F05.1 to F05.9. Kendall’s rank correlation coefficient was applied to test for trend.

RESULTS

From April 6th 1998 to April 5th 2025 there were 4,517,359 inpatient episodes with a diagnosis of delirium. The incident rate of inpatient delirium in adult patients increased from 2.5 to 16.6 per 10,000 admissions between 1998 to 2025 ($p<0.001$).

The median age of patient with delirium increased from 74.8 to 82.2 years ($p<0.001$) whilst length of stay decreased from 13.5 to 9.5 days ($p<0.001$). However, both the increase in mean age and reduction in median length of stay have reached a plateau since 2012 and 2009 respectively.

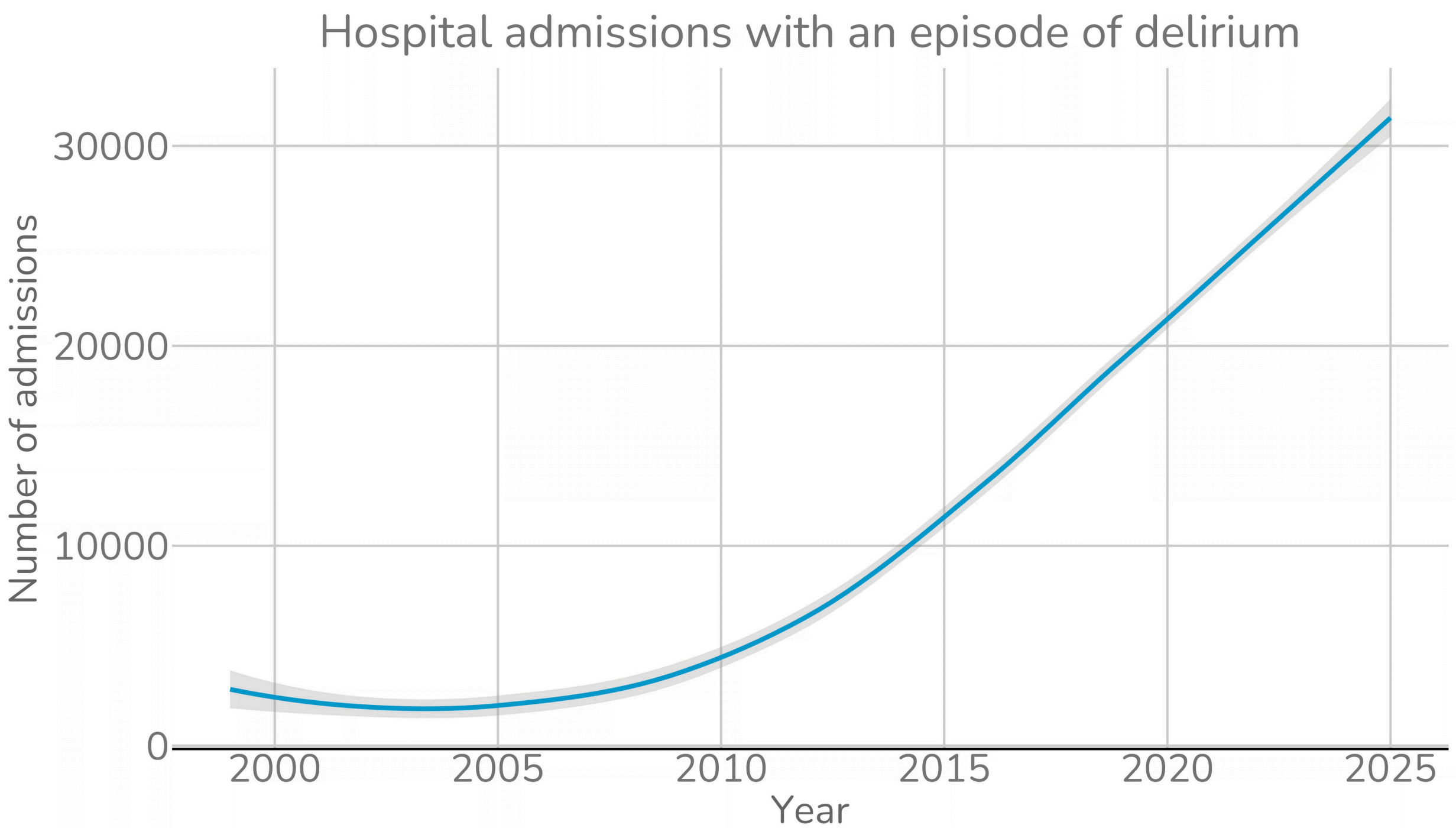


Figure 1. Number of hospital inpatients with a diagnosis of delirium (1998 to 2025).

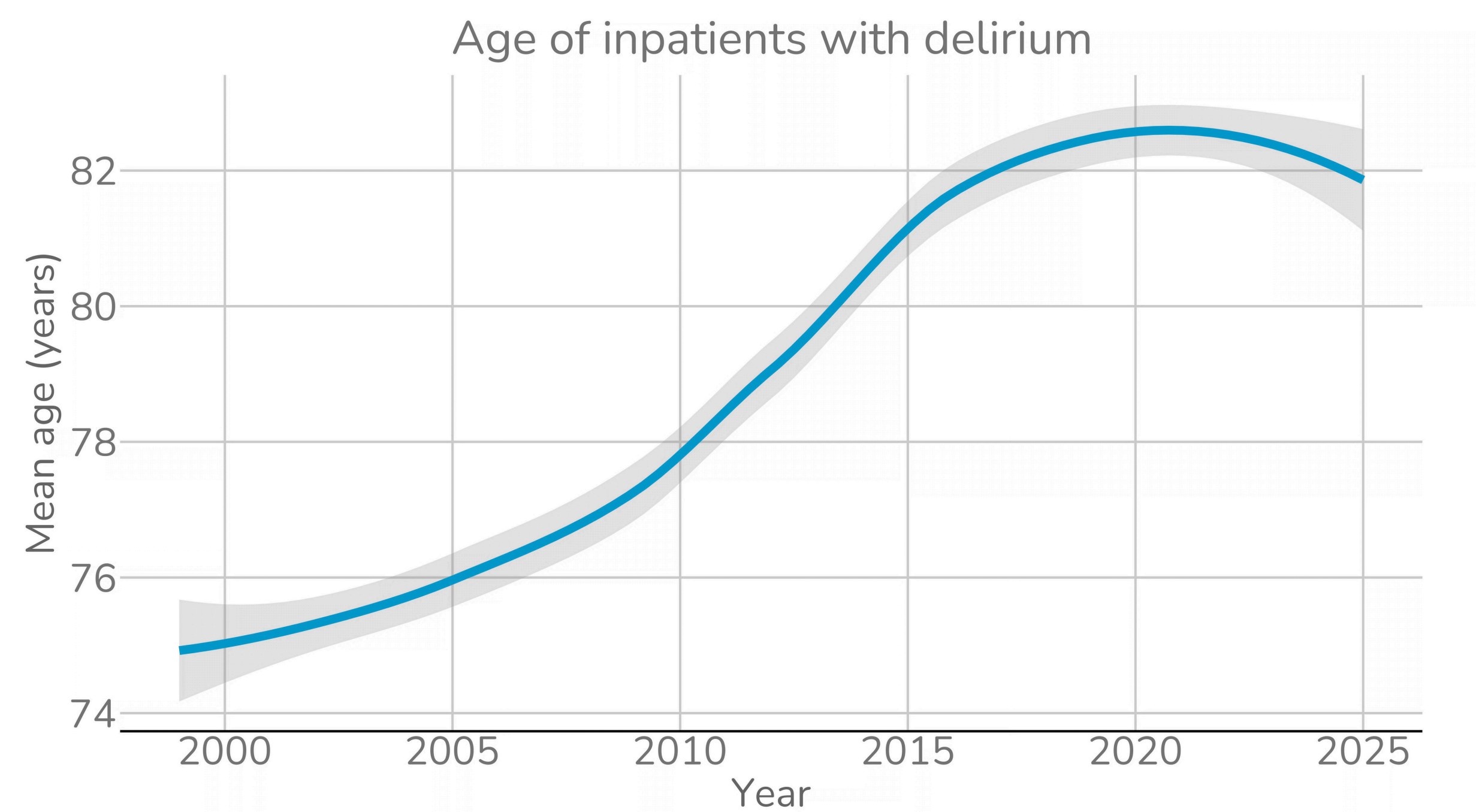


Figure 2. Mean age of hospital inpatients diagnosed with delirium between 1998 to 2025.

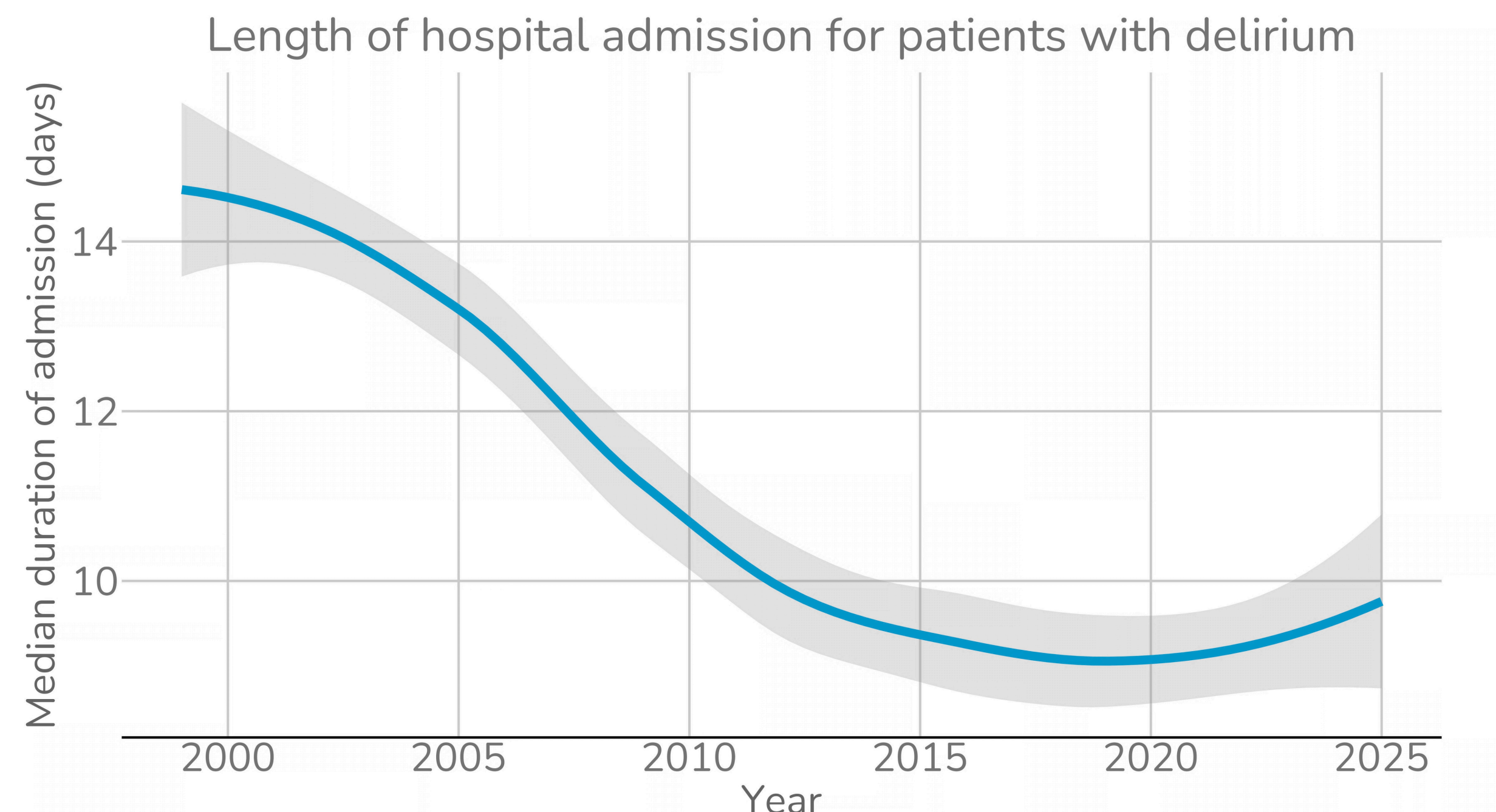


Figure 3. Median length of stay for hospital inpatients diagnosed with delirium between 1998 to 2025

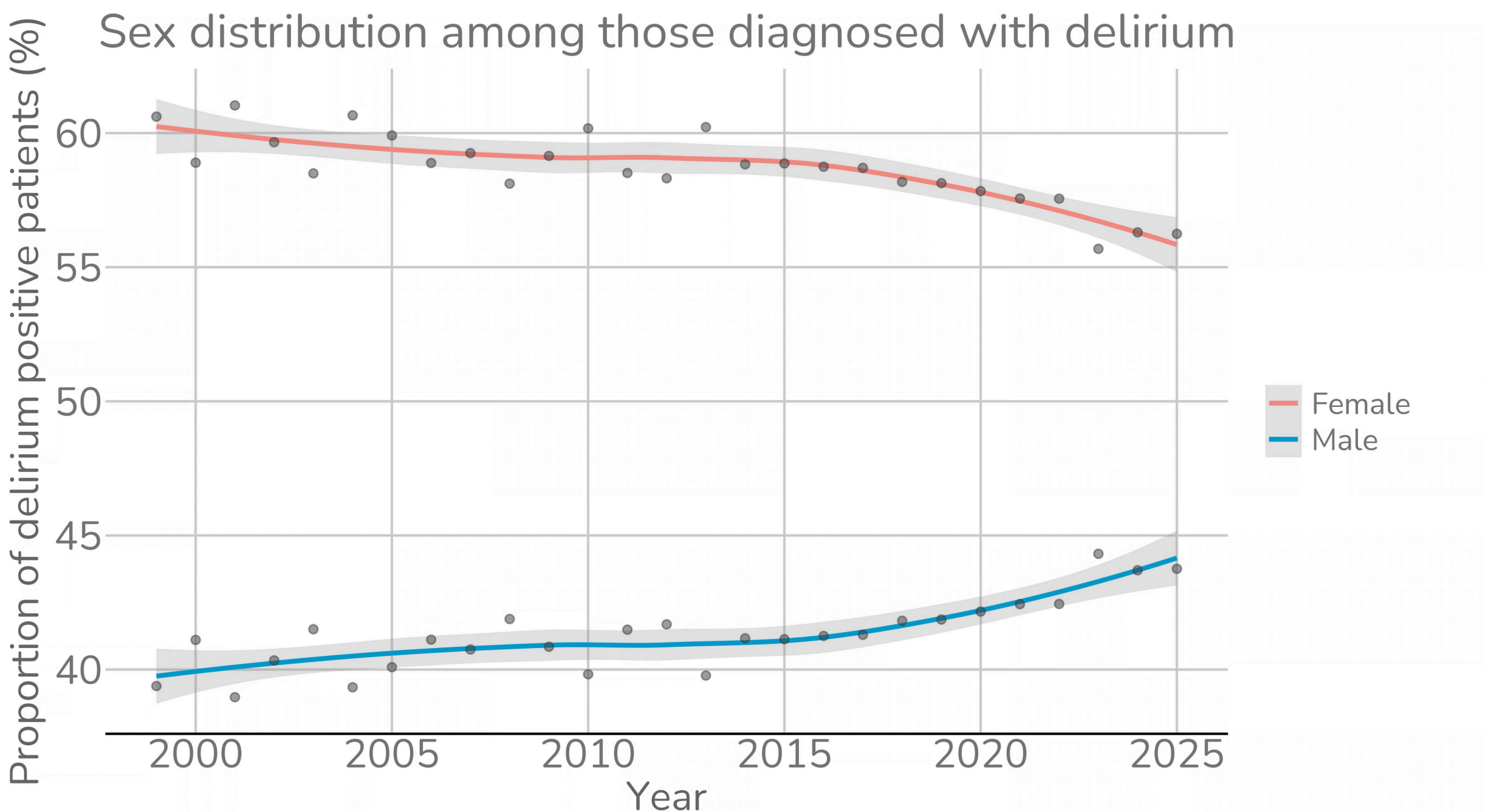


Figure 4. Sex distribution of hospital inpatients diagnosed with delirium between 1998 to 2025.

CONCLUSIONS

We found that the incidence of delirium is increasing which is expected based on increasing population age however the increase is in excess what would be expected from an aging population alone. This likely represents improved delirium screening and detection practices, consistent with individual hospital network studies comparing registry coding to active case identification (Pendlebury 2020). The increasing delirium incidence is consistent with our aging population structure. Interestingly, further improvements in admission duration reduction seen during 2000s have not continued over the past decade.

REFERENCES

Gordon et al. BMJ 2024 doi: 10.1136/bmj-2023-077634
Herbert et al. Int J Epidemiol 2017 doi:10.1093/ije/dyx015
Pendlebury et al. Clin Med. 2020 doi: 10.7861/clinmed.2019-0470

FUNDING

ML supported by an NIHR Doctoral Research Fellowship NIHR303253, and NIHR Oxford Biomedical Research Centre. The views expressed are those of the author(s) and not necessarily those of the NIHR or the Department of Health and Social Care.

CONTACT DETAILS

matthew.luney@univ.ox.ac.uk
ndorms.ox.ac.uk/team/matthew-luney

